

01 SCOPE & THE FLUID-MOSAIC MEMBRANE

Biochemistry = chemistry of life processes. In veterinary practice it supplies the **diagnostic** (serum enzymes, BUN), **nutritional** (limiting amino acid, EFA), **therapeutic** (fluid therapy, PGF_{2α}) and **toxicological** (xenobiotic, P450) framework.

Fluid mosaic model — **Singer & Nicolson, 1972**. A fluid lipid bilayer in which globular proteins float like icebergs. Bilayer = amphipathic phospholipids (polar head out, acyl tails in) + cholesterol + glycolipids. Proteins are **integral/intrinsic** (span bilayer, need detergent, e.g. Na⁺/K⁺-ATPase, glycoporphin) or **peripheral/extrinsic** (surface, salt extractable, e.g. spectrin, cyt c).

7–10 nm thickness	18% protein — myelin (lowest)	76% protein — inner mito (highest)	50:50 lipid:protein, RBC ghost
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- Asymmetry**: outer leaflet = phosphatidylcholine, sphingomyelin, **all glycolipids & glycoproteins**; inner leaflet = phosphatidylethanolamine, phosphatidylserine, PIP₂. **PS flipped outward = "eat-me" apoptosis signal** (basis of Annexin-V assay).
- Movement of lipids**: lateral diffusion & flexion are **fast**; transverse flip-flop is **very slow** and needs enzymes — **flippase** (inward, ATP), **floppase** (outward, ATP), **scramblase** (both ways, Ca²⁺-activated, no ATP).
- Fluidity** ↑ by unsaturated & short-chain fatty acids and ↑ temperature; ↓ by saturated/long chains. **Cholesterol is a fluidity buffer** — below transition temp it prevents crystalline packing (↑fluidity); above it restrains chains (↓fluidity).
- Functions**: selective permeability, compartmentation, electrical excitability, receptor & signal transduction, cell recognition (glycocalyx), antigenicity (blood groups), cell adhesion, energy transduction (mitochondrial inner membrane).

Trap: the bilayer is freely permeable to **small uncharged** molecules (O₂, CO₂, N₂, urea, ethanol, steroids). It is essentially impermeable to **ions and to glucose** — even Na⁺ and K⁺, being small, are blocked because of their large hydration shells.

02 TRANSPORT ACROSS MEMBRANES — THE MASTER MAP

MEMBRANE TRANSPORT

PASSIVE — down gradient, no ATP

ACTIVE — uphill, energy needed

Simple diffusion

no carrier · non-saturable
linear with gradient
O₂, CO₂, urea, steroids
+ aquaporins for H₂O

Facilitated diffusion

carrier · saturable (V_{max})
specific · inhibitable
GLUT 1–5, urea & anion
exchanger (band 3)

Primary active

ATP hydrolysed directly
Na⁺/K⁺-ATPase (3Na out:2K in)
Ca²⁺-ATPase, H⁺/K⁺-ATPase
P-, F-, V-, ABC types

Secondary active

rides the Na⁺ gradient
symport: SGLT-1 (2Na:1glc)
antiport: Na⁺/Ca²⁺, Na⁺/H⁺
stops if ATPase poisoned

Uniport one solute, one way — GLUT1

Symport two solutes, same way — SGLT-1

Antiport two solutes, opposite — Cl⁻/HCO₃⁻

Bulk endo/exo/pinocytosis, phagocytosis

Ionophores valinomycin (K⁺ carrier) · gramicidin (channel)

Most-asked: Na⁺/K⁺-ATPase — electrogenic (net +1 out), consumes **≈30 % of basal ATP**, inhibited by **ouabain / digitalis / cardiac glycosides** and by K⁺-free medium; it creates the Na⁺ gradient that powers **all** secondary active transport, and maintains cell volume against Donnan swelling.

03 GLUT FAMILY — ONE LINE EACH

CARRIER	SITE	POINT THAT SCORES
GLUT-1	RBC, brain, BBB	Low K _m ≈ 1 mM — basal uptake
GLUT-2	Liver, β-cell, kidney, gut basolateral	High K_m ≈ 15–20 mM = the glucose sensor ; bidirectional
GLUT-3	Neurons, placenta	Lowest K _m — neuron gets first claim
GLUT-4	Skeletal & cardiac muscle, adipose	Only insulin-dependent one — vesicle recruitment
GLUT-5	Intestinal brush border, sperm	Fructose transporter
SGLT-1/2	Gut / renal PCT	Secondary active, Na ⁺ -coupled; phlorizin inhibits

04 GIBBS-DONNAN MEMBRANE EQUILIBRIUM

Statement. When a **non-diffusible charged ion** (plasma protein anion, Pr⁻) is confined to one side of a semipermeable membrane, the diffusible ions redistribute **unequally** so that the **product of diffusible ion concentrations is equal on both sides**, while electroneutrality holds on each side.

$$[Na^+]_i \times [Cl^-]_i = [Na^+]_o \times [Cl^-]_o \cdot \text{Donnan ratio } r = [Na^+]_i/[Na^+]_o = [Cl^-]_o/[Cl^-]_i \approx 0.95$$

- Three consequences**: ① the protein side holds **more diffusible cations, fewer anions**; ② total osmotically active particles are higher on the protein side → water pulled in (**Donnan effect adds ≈ 1/3 extra to colloid osmotic pressure**); ③ a **Donnan potential** develops across the membrane.
- Veterinary significance**: plasma vs interstitial fluid ion split; contribution to resting membrane potential; the **chloride shift (Hamburger phenomenon)** in RBC; oedema of hypoproteinaemia (parasitism, PLE, liver failure) — falling albumin collapses both oncotic and Donnan pressure; behaviour of colloids (dextran, plasma) in fluid therapy; ultrafiltration in the glomerulus.
- Double Donnan / pump-leak**: if Donnan forces acted alone the cell would swell and burst. Na⁺/K⁺-ATPase keeps Na⁺ functionally impermeant outside, opposing it — cell volume is a **steady state, not equilibrium**.

05 DISSOCIATION OF ACIDS • PH • BUFFERS • HENDERSON-HASSELBALCH

Brønsted-Lowry: acid = proton donor, base = proton acceptor. Strength of a weak acid is its **pK_a** = pH at which it is exactly **half dissociated**. Lower pK_a = stronger acid. **pH = -Log[H⁺]**; blood [H⁺] ≈ 40 nmol/L.

Buffer = weak acid + its conjugate base; resists pH change on adding acid/alkali. **Buffer capacity is maximal at pH = pK_a** and useful only within **pK_a ± 1** — this is why the flat middle of a titration curve is the buffering zone.

Henderson-Hasselbalch $pH = pK_a + \log ([salt]/[acid]) = pK_a + \log ([A^-]/[HA])$
For blood: $pH = 6.1 + \log (HCO_3^- / 0.03 \times pCO_2) = 6.1 + \log (24/1.2) = 6.1 + \log 20 = 7.4$

The 20 : 1 ratio is the single most examined number in the unit. pH depends on the **ratio**, not absolute values — which is why compensation (renal HCO₃⁻ retention, respiratory CO₂ blow-off) restores pH by restoring the ratio.

BUFFER SYSTEM	PK _A	WHERE IT MATTERS
HCO ₃ ⁻ / H ₂ CO ₃	6.1	Chief ECF buffer — weak pK but an open system : lungs dump CO ₂ , kidney regenerates HCO ₃ ⁻
Haemoglobin	~7.9 / 6.7	Chief intra-erythrocyte buffer; deoxy-Hb is the better buffer → isohydric CO ₂ carriage
Protein (imidazole of His)	6.8	Main intracellular buffer; also plasma albumin
HPO ₄ ²⁻ / H ₂ PO ₄ ⁻	6.8	Intracellular + urinary titratable acidity ; bone reservoir
Ammonia (NH ₃ /NH ₄ ⁺)	9.2	Renal tubular acid excretion in chronic acidosis

7.35–7.45 arterial blood pH	24 mM HCO ₃ ⁻	40 mmHg pCO ₂	6.2–7.0 rumen pH (SARA < 5.6)	2–3 abomasum / gastric	6.6–6.8 milk (↑ in mastitis)
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Trap: herbivore urine is normally **alkaline** (7.5–8.5), carnivore urine **acidic** (5.5–7.0) — a "normal" pH 8 urine in a dog is abnormal (urease-positive infection).

Veterinary Biochemistry • Unit 1 • General Veterinary Biochemistry

MEMBRANE → TRANSPORT → DONNAN → BUFFERS

86 DEFINITION, CLASSIFICATION & ISOMERISM

Carbohydrates = polyhydroxy aldehydes or ketones, or substances yielding them on hydrolysis; empirical formula (CH₂O)_n. Classified as **monosaccharides** (cannot be hydrolysed), **disaccharides**, **oligosaccharides** (2–10), **polysaccharides** (homo- vs hetero-).

Aldose ~CHO (glucose)

Ketose C=O (fructose)

Epimers differ at ONE C: glucose↔galactose at C-4, glucose↔mannose at C-2

Anomers α/β at C-1 (anomeric carbon)

Enantiomers D/L — mirror at penultimate C; body sugars are D

Mutarotation α +112° → +52.7° → β +18.7°

- Reducing sugars** have a free anomeric –OH → reduce Benedict's/Fehling's. **All monosaccharides**, maltose, isomaltose, lactose, cellobiose are reducing. **Sucrose and trehalose are non-reducing** (both anomeric carbons locked).
- Ring forms:** pyranose (6-membered) & furanose (5-membered); Haworth projection; chair form of β-D-glucopyranose is the most stable sugar because **all bulky groups are equatorial**.
- Derivatives:** sugar acids (gluconic, glucuronic — conjugation & detoxification, ascorbic acid), sugar alcohols (sorbitol → cataract in diabetes, glycerol, inositol), deoxy sugars (deoxyribose, fucose), **amino sugars** (glucosamine, galactosamine, **N-acetylneuraminic acid = sialic acid**), sugar esters (glucose-6-P).

87 MONOSACCHARIDES — BIOLOGICAL SIGNIFICANCE

SUGAR	WHY IT EXISTS IN THE ANIMAL
Ribose (deoxyribose)	Backbone of RNA / DNA , ATP, GTP, NAD ⁺ , FAD, CoA, SAM. Sole source = HMP shunt . Ribose-5-P is the entry point of nucleotide synthesis.
Glucose	"Blood sugar", universal fuel; the only fuel for brain, RBC, renal medulla under normal conditions. Stored as glycogen, converted to fat.
Fructose	Ketohexose, sweetest sugar ; enters glycolysis independent of insulin. Chief energy source of spermatozoa in bull/ram semen (fructolysis — a semen-quality test); high in fetal blood of ruminants.
Galactose	Made in the udder from glucose → lactose synthesis ; component of cerebrosides, gangliosides, glycoproteins, blood-group antigens. Galactosaemia if metabolism blocked.
Mannose	Core sugar of N-linked glycoproteins ; mannose-6-phosphate is the address tag that routes enzymes to lysosomes (I-cell disease when lost).
Amino sugars	Glucosamine & galactosamine → GAGs, cartilage, chitin; sialic acid caps glycoproteins & determines RBC lifespan and virus attachment (influenza).

Species hook: ruminant blood glucose is only **45–60 mg/dL** (dog/cat 70–110) because dietary carbohydrate is fermented to VFA in the rumen — the ruminant is a **gluconeogenic animal living on propionate**. This single fact explains ketosis, pregnancy toxemia and the whole of Unit 3.

88 DISACCHARIDES — LINKAGE IS THE ANSWER

SUGAR	UNITS & LINKAGE	RED.	NOTE
Maltose	Glc α1→4 Glc	Yes	Malt sugar; from starch by amylase
Isomaltose	Glc α1→6 Glc	Yes	From branch points of amylopectin
Lactose	Gal β1→4 Glc	Yes	Milk sugar , only in mammary gland; osmotic regulator of milk volume
Sucrose	Glc α1→2β Fru	No	Cane sugar; hydrolysis → invert sugar ; no free anomeric C
Cellobiose	Glc β1→4 Glc	Yes	From cellulose ; only rumen/hindgut microbes split it
Trehalose	Glc α1→1α Glc	No	"Insect blood sugar"; fungal

4.7%
lactose, cow milk

4.9%
buffalo

6.2%
mare — highest

≈0
lactase in adult cat/dog

Trap: lactose is the **only** sugar of purely animal origin. Adult ruminants and most adult carnivores lose intestinal lactase — feeding milk causes osmotic diarrhoea, a classic viva point.

89 POLYSACCHARIDES — HOMOGLYCANS

POLYSACCHARIDE	MONOMER & LINKAGE	BRANCHING	VETERINARY SIGNIFICANCE
Starch — amylose	Glucose, α1→4	Unbranched helix	Stains deep blue with iodine; plant storage
Starch — amylopectin	Glucose, α1→4 + α1→6	Every 24–30 residues	Stains violet/purple ; 80 % of starch
Dextrins	Partial hydrolysis products	—	Erythrodextrin (red with I ₂) → achrodextrin (no colour); limit dextrin resists amylase
Dextran	Glucose, α1→6 chief	α1→3, α1→4	Bacterial; plasma volume expander (colloid) ; dental plaque matrix
Glycogen	Glucose, α1→4 + α1→6	Every 8–12 residues	Animal storage : liver 5–6 % (blood glucose homeostasis), muscle 0.7–1 % (local fuel only — no glucose-6-phosphatase). Dense branching = many non-reducing ends = rapid mobilisation. Stains red-brown with iodine.
Cellulose	Glucose, β1→4	Linear, H-bonded fibrils	No mammal makes cellulase ; rumen/caecal microbes ferment it → VFA. Basis of herbivore nutrition & crude fibre.
Inulin	Fructose, β2→1	Linear	Freely filtered, not reabsorbed/secreted → gold-standard GFR marker ; prebiotic
Chitin	N-acetylglucosamine, β1→4	Linear	Arthropod exoskeleton & fungal cell wall — target of antifungals; second most abundant polysaccharide

One-line contrast: Starch, glycogen = α-linked = digestible by mammalian amylase. Cellulose, chitin = β-linked = need microbial enzymes. Digestibility is decided by the anomeric configuration, not by the sugar.

10 MUCOPOLYSACCHARIDES (GAGS) & BACTERIAL CELL-WALL POLYSACCHARIDES

GAGs = long unbranched heteropolysaccharides of a repeating disaccharide = **uronic acid (or galactose) + amino sugar**, almost all **sulphated** and strongly anionic. Bound to a core protein they form **proteoglycans** (ground substance) — a huge hydrated gel that resists compression, lubricates and traps cations and water.

GAG	REPEATING UNIT	SITE / USE
Hyaluronic acid	GlcUA + GlcNAc — only non-sulphated, no protein core	Synovial fluid, vitreous humour, Wharton's jelly, loose CT. Hyaluronidase = "spreading factor" of sperm acrosome & invasive bacteria
Chondroitin 4-/6-sulphate	GlcUA + GalNAc-SO ₄	Most abundant GAG — cartilage, bone, cornea; joint supplements
Keratan sulphate	Galactose + GlcNAc-SO ₄ (no uronic acid)	Cornea (transparency), cartilage, intervertebral disc
Dermatan sulphate	IdUA + GalNAc-SO ₄	Skin, blood vessels, heart valves
Heparin	IdUA-SO ₄ + GlcN-SO ₄	Most acidic / highest negative charge ; mast-cell granules; anticoagulant (antithrombin III); releases lipoprotein lipase — "clearing factor"
Heparan sulphate	Similar, less sulphated	Cell surface, basement membrane — glomerular charge barrier (loss → proteinuria)

Clinical: deficiency of lysosomal hydrolases → **mucopolysaccharidoses** (accumulation in cornea, bone, brain; reported in cats and dogs — MPS I, VI, VII). Corneal clouding + skeletal dysplasia + dwarfism is the classic triad.

Bacterial cell wall — peptidoglycan (murein). Alternating **N-acetylglucosamine (NAG)** and **N-acetylmuramic acid (NAM)** joined β1→4; NAM carries a tetrapeptide, and chains are cross-linked by **transpeptidase** (pentaglycine bridge in *Staphylococcus*).

NAG-NAM β1→4 → tetrapeptide → transpeptidase cross-link → rigid sacculus

- Lysozyme** (tears, saliva, egg white, phagocytes) hydrolyses the **β1→4 NAM-NAG** bond → lysis. First-line innate defence.
- Penicillins & cephalosporins** inhibit transpeptidase (PBP) → wall cannot cross-link. Selective toxicity rests on the fact that **the host has no peptidoglycan**.
- Gram-positive:** thick peptidoglycan + **teichoic acid** (glycerol/ribitol phosphate — adhesion, antigenicity).
- Gram-negative:** thin peptidoglycan + outer membrane bearing **lipopolysaccharide (LPS)**: O-antigen + core + **lipid A = the endotoxin** — the trigger of coliform mastitis, endotoxic shock and DIC (ties directly into Unit 3 "oxidative stress & shock").
- Capsular polysaccharides** resist phagocytosis and are the basis of serotyping and of many vaccines.

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BLOOR'S CLASSIFICATION — THE SKELETON OF EVERY LIPID ANSWER

LIPIDS (Bloor)

SIMPLE — ester of FA + alcohol

- Fats & oils = triacylglycerol (FA + glycerol) — fat is solid, oil liquid at room temp
- Waxes = FA + long-chain alcohol

COMPOUND — + extra group

- Phospholipids (glycero- / sphingo-)
- Glycolipids (cerebroside, ganglioside)
- Lipoproteins (CM, VLDL, LDL, HDL)
- Sulpholipids, amino-lipids

DERIVED — hydrolysis products

- Fatty acids, glycerol, mono/di-AG
- Steroids & sterols (cholesterol)
- Ketone bodies, fat-soluble vit ADEK
- Eicosanoids, terpenes, carotenoids

Common properties: insoluble in water, soluble in organic solvents (ether, chloroform, benzene); **amphipathic** if a polar head exists; highest calorific value **9.3 kcal/g** (vs 4.1 for carbohydrate/protein); non-polar so stored anhydrous — the most economical fuel store. **Functions:** concentrated energy, structural (membranes, myelin), thermal insulation & padding, carrier of vitamins ADEK, precursor of steroid hormones & eicosanoids, source of essential fatty acids, surfactant, milk-fat and egg-yolk quality.

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FATTY ACIDS & ESSENTIAL FATTY ACIDS

- Saturated:** butyric C4 (rumen VFA, milk fat), palmitic C16, stearic C18. **Unsaturated:** oleic 18:1 ω9, **linoleic 18:2 ω6**, **α-linolenic 18:3 ω3**, arachidonic 20:4 ω6, EPA 20:5, DHA 22:6.
- Essential fatty acids = linoleic + α-linolenic** — mammals lack Δ12 and Δ15 desaturase so cannot insert double bonds beyond C-9. **Arachidonic acid is semi-essential** (made from linoleic).
- EFA deficiency:** scaly dermatitis, dull dry coat, poor wound healing, impaired reproduction, fatty liver, ↑ transepidermal water loss.
- Cat exception:** the cat has negligible Δ6-desaturase → **arachidonic acid is dietary-essential for cats**, which is why cats are obligate carnivores nutritionally.
- Natural cis** configuration bends the chain (↑fluidity, ↓melting point); industrial **trans** behaves like saturated.

Trap: "ω" numbering counts from the **methyl** end, "Δ" from the **carboxyl** end. Arachidonic acid = 20:4 Δ5,8,11,14 = ω6.

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FAT INDICES — PURE OBJECTIVE MARKS

INDEX	MEASURES	VALUE / USE
Saponification no.	mg KOH to saponify 1 g fat	Inversely α chain length. Butter 210–230 (short chains), groundnut oil ≈190
Iodine no.	g I ₂ absorbed per 100 g fat	Degree of unsaturation. Coconut ≈8, butter 26–38, linseed 175–200
Acid no.	mg KOH to neutralise free FA in 1 g	Index of hydrolytic rancidity — the practical you perform
Reichert–Meissl no.	Volatile, water-soluble FA (butyric)	Butter 24–34; vanaspati <1 → test of ghee adulteration
Polenske no.	Volatile, water-insoluble FA	High in coconut oil — detects coconut-oil adulteration
Acetyl no.	Free –OH groups	High in castor oil (ricinoleic acid)

Rancidity. **Hydrolytic** = lipase/moisture → free FA, ↑acid number. **Oxidative** = O₂ attacks double bonds → peroxides, aldehydes; worst in ω-3-rich fish oil. Prevented by **vitamin E, BHT/BHA**. **Feed link:** rancid fish-oil diet + vitamin E deficiency → **yellow fat disease (steatitis / pansteatitis)** in cats and mink.

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COMPOUND LIPIDS — THE ONES EXAMINERS NAME

LIPID	IDENTITY & ROLE
Lecithin (PC)	Choline; most abundant phospholipid; emulsifier ; dipalmitoyl-lecithin = lung surfactant ; egg yolk
Cephalin (PE)	Ethanolamine; brain; role in blood coagulation (thromboplastin)
Phosphatidyl-serine	Inner leaflet; externalised in apoptosis ; procoagulant surface
Phosphatidyl-inositol	PIP₂ → IP₃ + DAG: the Ca ²⁺ second-messenger system; GPI membrane anchors
Cardiolipin	Diphosphatidylglycerol — found only in the inner mitochondrial membrane ; antigenic (VDRL antigen)
Plasmalogens	Ether linkage at C-1; abundant in heart & muscle , myelin
Sphingomyelin	Only phospholipid with sphingosine, not glycerol ; myelin sheath; Niemann–Pick disease
Cerebrosides	Ceramide + one sugar (galactose); white matter; Gaucher/Krabbe
Gangliosides	Ceramide + oligosaccharide + sialic acid ; GM1 is the cholera-toxin receptor ; Tay–Sachs
Cholesterol	27-C, cyclopentanoperhydrophenanthrene ring, exclusively animal; precursor of bile acids, vitamin D₃, all steroid hormones , membrane fluidity buffer

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LIPOPROTEINS — DENSITY RISES AS FAT FALLS

apo B-48

apo B-100

apo B-100

apo A-I

CM

VLDL

LDL

HDL

largest · least dense · most TG

smallest · densest · most protein

CLASS	CARRIES	FROM → TO
Chylomicron	Exogenous TG (dietary)	Intestine → tissues; lymph/lacteals
VLDL	Endogenous TG	Liver → tissues; failure to export → fatty liver
LDL	Cholesterol ("bad")	→ peripheral tissue, receptor-mediated endocytosis
HDL	Cholesterol ("good")	Reverse cholesterol transport , tissue → liver; activates LCAT

- Lipoprotein lipase** on capillary endothelium hydrolyses TG in CM/VLDL; activated by **apo C-II**, released into blood by **heparin** ("clearing factor lipase") — the reason post-prandial plasma is lipaemic/milky.
- Species point:** dog, cat and most farm species are **"HDL animals"** (HDL predominant) → spontaneous atherosclerosis is rare; hypercholesterolaemia in dogs points to **hypothyroidism, diabetes, nephrotic syndrome, cholestasis or hyperadrenocorticism**.

16

PROSTAGLANDINS & THE EICOSANOID TREE

Eicosanoids = 20-carbon (*eicosa*) derivatives of **arachidonic acid**, released from membrane phospholipid by **phospholipase A₂**. They are **local hormones (autacoids)** — made on demand, act on the cell of origin or its neighbour, and are destroyed within seconds; they are **not stored**. Basic skeleton = prostanoid acid (cyclopentane ring + two side chains).

Membrane phospholipid

→ phospholipase A₂ ⊥ corticosteroids

→ Arachidonic acid

cyclo-oxygenase ⊥ NSAIDs / aspirin

→ PGG₂ → PGH₂

→ PGD/E/F, PGI₂, TXA₂

lipoxygenase

→ HPETE

→ Leukotrienes (LTB₄, LTC₄/D₄/E₄ = SRS-A)

- Nomenclature:** PG + letter (A–I, by ring substituent) + **subscript number = number of double bonds in the side chains**. Series 1 from dihomo-γ-linolenic, **series 2 from arachidonic (the important ones)**, series 3 from EPA — which is why fish-oil ω-3 gives **less inflammatory** eicosanoids.
- Aspirin** irreversibly acetylates COX (platelets cannot resynthesise it → long antiplatelet effect); other **NSAIDs are reversible**; **corticosteroids block phospholipase A₂** and therefore switch off **both** arms — the reason steroids are broader anti-inflammatories than NSAIDs.

EICOSANOID	CHIEF SOURCE	PRINCIPAL ACTION
PGF_{2α}	Uterine endometrium	Luteolysis + myometrial contraction — the basis of oestrus synchronisation, induction of parturition, treatment of pyometra & retained placenta (dinoprost, cloprostenol)
PGE₂	Most tissues	Vasodilation, fever & pain sensitisation , gastric mucus/bicarbonate cytoprotection , bronchodilation, cervical relaxation
PGI₂ (prostacyclin)	Vascular endothelium	Vasodilation, inhibits platelet aggregation
TXA₂	Platelets	Vasoconstriction, promotes platelet aggregation — exact opposite of PGI ₂
LTB₄	Leucocytes	Chemotaxis of neutrophils
LTC₄/D₄/E₄	Mast cells	SRS-A: bronchoconstriction, ↑vascular permeability — anaphylaxis, equine asthma

The **PGI₂ → TXA₂ balance** (endothelium vs platelet) governing haemostasis, and **PGF_{2α} luteolysis**, are the two prostaglandin facts most likely to be asked in both objective and long-answer form.

Trap: NSAID toxicity in dogs (gastric ulceration, renal papillary necrosis) is not a side reaction — it is the **predictable loss of PGE₂ cytoprotection and PGI₂-dependent renal perfusion**.

Veterinary Biochemistry · Unit 1 · Lipids

PLA₂ → COX / LOX → LOCAL HORMONES

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CLASSIFICATION OF PROTEINS

GROUP	MEMBERS & IDENTIFIERS
SIMPLE (yield only AA)	Albumin — water soluble, coagulable, full-saturation (NH ₄) ₂ SO ₄ ; serum albumin, ovalbumin, lactalbumin. Globulin — insoluble in water, soluble in dilute salt, half-saturation salting-out; immunoglobulins. Glutelin (glutenin), Prolamin (zein of maize, gliadin — deficient in lysine & tryptophan). Histone — basic, arginine/lysine rich, bound to DNA. Protamine — most basic, in fish sperm. Albuminoid / scleroprotein — most insoluble : collagen, elastin, keratin.
CONJUGATED (+ prosthetic group)	Nucleoprotein (nucleic acid) · Glycoprotein / mucoprotein (CHO — mucin, hormones) · Phosphoprotein (casein, ovovitellin) · Chromoprotein (haem — Hb, myoglobin, cytochromes, catalase) · Lipoprotein · Metalloprotein (ceruloplasmin—Cu, ferritin—Fe, carbonic anhydrase—Zn) · Flavoprotein (FAD).
DERIVED	Primary (bonds intact): proteans, metaproteins, coagulated proteins. Secondary (progressive hydrolysis): protein → protease → peptone → polypeptide → peptide → amino acid .

Functional list for a 10-mark "biological significance" answer: structural (collagen, keratin) · catalytic (enzymes) · transport (Hb, albumin, transferrin, ceruloplasmin) · hormonal (insulin, GH) · defence (immunoglobulins, complement) · clotting (fibrinogen) · contractile (actin, myosin) · **colloid osmotic pressure (albumin ≈ 80 %)** · buffering · storage (casein, ferritin, ovalbumin) · genetic (histones) · nutrition (10 essential AA) · receptors & signalling.

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PROPERTIES OF PROTEINS

- Amphoteric & isoelectric point (pI)**: net charge zero at pI → minimum solubility, maximum precipitation, no movement in an electric field. **Casein pI = 4.6** — the reason milk curdles when it sours or when acid/rennet is added, and the basis of paneer/dahl. Serum albumin pI ≈ 4.9.
- Electrophoresis**: at pH 8.6 all plasma proteins are anionic and separate as **albumin → α₁ → α₂ → β → γ**. This is the direct link to **dysproteinemias** in Unit 3.
- Denaturation**: heat, extremes of pH, urea, guanidine, heavy metals, organic solvents, UV, vigorous shaking. **Secondary, tertiary and quaternary structure are destroyed; the peptide bonds of the primary structure remain intact**. Result: loss of biological activity, ↓solubility, ↑viscosity, ↑digestibility, exposure of –SH groups. Usually irreversible (boiled egg) but sometimes reversible (ribonuclease).
- Precipitation**: by heavy-metal ions (Hg²⁺, Pb²⁺ — protein/egg white is a first-aid antidote), by alkaloidal reagents (tungstic, trichloroacetic, picric, sulphosalicylic acid — used for deproteinising blood before glucose estimation), by organic solvents, and by **salting out** — globulin at half saturation and albumin at full saturation with ammonium sulphate.
- Colloidal nature**: large size, Tyndall effect, non-dialysable, exert osmotic (oncotic) pressure ≈ 25 mmHg.

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PHYSICAL & CHEMICAL PROPERTIES OF AMINO ACIDS · THE PEPTIDE BOND

Low pH
–NH₃⁺ / –COOH
cation — cathode

→

pI
zwitterion
no migration

→

High pH
–NH₂ / –COO[–]
anion — anode

Isoelectric pH: for a neutral amino acid $pI = (pK_1 + pK_2)/2$; for acidic and basic amino acids, average the two pK values that flank the zwitterion. Solubility and buffering are **minimum** at pI.

Optical activity. The α-carbon is asymmetric (except glycine) → D and L forms, dextro (+) or laevo (–) rotation. **Proteins contain only L-amino acids**; D-forms occur in bacterial cell walls (D-alanine, D-glutamate) and some antibiotics — which is why they resist host proteases.

Peptide bond formation. The –COOH of one amino acid condenses with the –NH₂ of the next with loss of water, giving a **substituted amide (–CO–NH–)** link. Chains are written N-terminal → C-terminal.

- Has **partial double-bond character** from resonance → the six atoms of the bond are **rigid and planar**; **no free rotation about the C–N bond**.
- Almost always **trans** (except X–proline); rotation is possible only about the **φ (phi) and ψ (psi)** bonds of the α-carbon — the basis of the **Ramachandran plot**.
- Uncharged but polar; the C=O and N–H are exactly the H-bond donors/acceptors that build the α-helix and β-sheet.
- Hydrolysed by strong acid/alkali or by proteolytic enzymes; two or more peptide bonds give the **biuret reaction**.

Trap: Biuret is negative for free amino acids and for dipeptides — it needs at least two peptide bonds (i.e. a tripeptide onwards).
Ninhydrin, in contrast, is positive for free α-amino acids.

REACTION	DETECTS	COLOUR
Ninhydrin	Free α-amino acids	Blue-violet (Ruhemann's purple); proline → yellow
Biuret	≥ 2 peptide bonds	Violet / pink
Millon's	Tyrosine (phenol)	Brick red
Xanthoproteic	Aromatic ring	Yellow → orange with alkali
Hopkins–Cole	Tryptophan (indole)	Violet ring
Sakaguchi	Arginine (guanidino)	Red
Lead acetate	Cysteine (–SH)	Black PbS
Sanger's (FDNB)	N-terminal AA	Yellow DNP-AA
Edman (PITC)	N-terminal, sequential	PTH-amino acid

Sørensen's formol titration (your practical): formaldehyde blocks the –NH₂ group, removing its basic character, so the –COOH can be titrated against standard NaOH with phenolphthalein — this is how amino acids are **estimated**, and it is a favourite "principle" question.

22 BASES → NUCLEOSIDE → NUCLEOTIDE

BASE
purine / pyrimidine

+

PENTOSE
ribose / deoxyribose

=

NUCLEOSIDE
N-glycosidic bond

+

P

NUCLEOTIDE
ester at 5'-OH

N-9 of purine or N-1 of pyrimidine joins C-1' of the sugar · β-configuration

Purines Fused 6+5 ring. **Adenine, Guanine**; also hypoxanthine, xanthine, **uric acid** (end product in birds & reptiles = uricotelic; in Dalmatian dogs), and methylxanthines **caffeine & theobromine** — theobromine of chocolate is the classic **canine toxicosis**.

Pyrimidines Single 6-ring. **Cytosine (both), Thymine (DNA only), Uracil (RNA only)**; orotic acid is the pathway intermediate. Mnemonic: **CUT the PY** — Cytosine, Uracil, Thymine are PYrimidines.

Naming Adenine → adenosine → AMP/ADP/ATP; Guanine → guanosine → GMP; Cytosine → cytidine; Uracil → uridine; Thymine → thymidine. Add "deoxy" for the DNA series.

Biological significance of nucleotides (a guaranteed short note): ① **energy currency** — ATP, GTP, and the high-energy phosphoanhydride bond; ② **second messengers** — cAMP, cGMP; ③ **coenzymes** — NAD⁺, NADP⁺, FAD, CoA, all built on an **adenine nucleotide**; ④ **activated carriers** — UDP-glucose (glycogen), CDP-choline (phospholipid), **SAM** (methyl), PAPS (sulphate, for GAGs & detoxification); ⑤ **monomers** of DNA & RNA; ⑥ allosteric regulators (ATP/AMP ratio); ⑦ drug targets — allopurinol, 5-fluorouracil, azathioprine.

24 RNA — A TYPICAL RIBONUCLEIC ACID

Single-stranded, **ribose** sugar, **uracil in place of thymine**, alkali-labile (2'-OH), can fold on to itself into local double-helical hairpins.

TYPE	%	STRUCTURE & ROLE
mRNA	2–5	Most heterogeneous, shortest-lived. Eukaryotic mRNA is capped with 7-methylguanosine at 5' and carries a 3' poly-A tail ; carries codons — the working copy of the gene
tRNA	≈15	Smallest RNA; clover-leaf secondary / inverted-L tertiary shape; has an anticodon arm and a 3'-CCA end that accepts the amino acid ; rich in unusual bases (pseudouridine, dihydrouridine, inosine). Adaptor molecule
rRNA	≈80	Most abundant ; with proteins forms the ribosome — prokaryote 70S (50S + 30S), eukaryote 80S (60S + 40S) ; the peptidyl transferase is a ribozyme . Target of aminoglycosides & macrolides
Others	—	snRNA (splicing), miRNA/siRNA (gene silencing), hnRNA (mRNA precursor)

Trap that costs marks: "smallest" = **tRNA**; "most abundant" = **rRNA**; "most variable / shortest half-life" = **mRNA**. Examiners rotate these three every year.

25 FINAL RECALL BANK — READ THESE LAST, STRAIGHT BEFORE THE PAPER

Fluid mosaic model proposed by	Singer & Nicolson, 1972	Energy value of fat	9.3 kcal/g
Membrane with highest lipid / lowest protein	Myelin	Essential fatty acids	Linoleic & α-linolenic
Membrane with highest protein	Inner mitochondrial membrane	EFA additionally essential for the cat	Arachidonic acid
Enzyme moving lipid inward / outward	Flippase / floppase	Index of unsaturation	Iodine number
Only insulin-dependent glucose transporter	GLUT-4	Index of rancidity	Acid number
Glucose sensor of β-cell & liver	GLUT-2 (high K _m)	Test for ghee adulteration	Reichert-Meissl number
Inhibitor of Na ⁺ /K ⁺ -ATPase	Ouabain / digitalis	Saponification number is <	1 / fatty-acid chain length
Na ⁺ -glucose symporter inhibitor	Phlorizin	Phospholipid of lung surfactant	Dipalmitoyl lecithin
Donnan equation	[Na] _i [Cl] _i = [Na] _o [Cl] _o	Phospholipid only in inner mito membrane	Cardiolipin
Buffer works best at	pH = pK _a (± 1)	Only phospholipid without glycerol	Sphingomyelin
pK _a of bicarbonate system	6.1	Largest, least dense lipoprotein	Chylomicron
HCO ₃ ⁻ : H ₂ CO ₃ ratio in blood	20 : 1	Reverse cholesterol transport	HDL (apo A-I, LCAT)
Chief intracellular buffer	Protein (imidazole) & phosphate	Activator of lipoprotein lipase	Apo C-II (released by heparin)
Chief buffer inside the RBC	Haemoglobin	Ring system of cholesterol	Cyclopentanoperhydrophenanthrene
Epimers of glucose at C-4 / C-2	Galactose / mannose	Precursor of all eicosanoids	Arachidonic acid (20 C)
Specific rotation of D-glucose	+52.7°	Enzyme blocked by corticosteroids	Phospholipase A ₂
Sweetest sugar	Fructose	Enzyme blocked by aspirin/NSAIDs	Cyclo-oxygenase
Energy source of spermatozoa	Fructose	Prostaglandin causing luteolysis	PGF _{2α}
Non-reducing disaccharides	Sucrose, trehalose	Platelet aggregator vs inhibitor	TXA ₂ vs PGI ₂
Milk sugar & its linkage	Lactose, Gal β1–4 Glc	Most basic protein	Protamine
Glycogen branch frequency	Every 8–12 residues (α1–6)	Phosphoprotein of milk & its pl	Casein, pl 4.6
Polysaccharide used to measure GFR	Inulin	Residues per turn of α-helix	3.6 (pitch 5.4 Å)
Only non-sulphated GAG	Hyaluronic acid	Repeating unit of collagen	Gly–X–Y (Y = hydroxyproline)
Most acidic GAG / anticoagulant	Heparin	Bond retained after denaturation	Peptide bond (primary structure)
"Spreading factor" enzyme	Hyaluronidase	Only optically inactive amino acid	Glycine
Bacterial wall polymer & its sugars	Peptidoglycan — NAG + NAM	Purely ketogenic amino acids	Leucine & lysine
Enzyme in tears that lyses bacteria	Lysozyme (β1–4 NAM-NAG)	Amino acids absorbing at 280 nm	Phe, Tyr, Trp
Endotoxin of Gram-negative wall	Lipid A of LPS	Ninhydrin colour with proline	Yellow (others blue-violet)
		Method for estimating amino acids	Sørensen's formol titration
		Hydrogen bonds in A=T / G≡C	2 / 3
		Base pairs per turn & pitch of B-DNA	10 bp, 3.4 nm
		Left-handed DNA	Z-DNA
		Increase in A ₂₆₀ on melting	Hyperchromic effect
		Smallest / most abundant RNA	tRNA / rRNA
		Amino-acid attachment site on tRNA	3'-CCA end

Thirty-second closing sweep: 1972 Singer–Nicolson · 3 Na out : 2 K in · [Na][Cl] products equal · pH = pK_a + log salt/acid, ratio 20:1 · α = digestible, β = microbial · branch every 8–12 in glycogen · hyaluronic acid alone unsulphated · iodine = unsaturation, acid = rancidity, RM = ghee · PLA₂ → COX → PGF_{2α} lyses the CL · 3.6 residues per helical turn · PVT TIM HALL · Leu & Lys are the only ketogenic pair · A=T two bonds, G≡C three · 10 bp, 3.4 nm, right-handed B-DNA.